

Arithmetic Foundation: Powers, Exponents & Repeated Multiplication
Exam (Set 6)

Name:	Class:	Student ID:	Time: 30 Mins	Total: 100 Pts	Score:
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Section 1: Write it Out! Expanded Form vs Simple Multiplication (3 pts each, Score: _____
24 pts)

Write out the expanded multiplication form and calculate the result:

Expression	Expanded Form (e.g. $a \times a \times \dots$)	True Value	Common Slip (DO NOT MAKE)
2^3	$2 \times 2 \times 2$	8	$2 \times 3 = 6$ ✗
2^4			$2 \times 4 = 8$ ✗
3^3			$3 \times 3 = 9$ ✗
3^4			$3 \times 4 = 12$ ✗
4^2			$4 \times 2 = 8$ ✗
5^3			$5 \times 3 = 15$ ✗
10^3			$10 \times 3 = 30$ ✗

Section 2: Addition of Exponents vs Multiplication of Exponents (4 pts each, Score: _____
24 pts)

1. Simplify each expression into a single power (State why exponents are added):

(a) $10^4 \times 10^5 = 10^{+} =$ _____	(b) $2^3 \times 2^4 = 2^{+} =$ _____
(c) $3^2 \times 3^5 = 3^{+} =$ _____	(d) $10^0 =$ _____
(e) $5^1 \times 5^2 \times 5^3 = 5^{+} =$ _____	(f) $10^{-3} = \frac{1}{10^{+}} =$ _____

Section 3: Factors vs Multiples & Square Numbers Concept (4 pts each, 24 Score: _____
pts)

2. [Factors of 16] A student claims: "16 can be written as eight 2s."

(a) Is the student thinking about addition or multiplication?

(b) What is the true prime factor decomposition of 16 in index notation?

(a) The student is confusing addition with multiplication: $8 \times 2 = 16$ is $2 + 2 + 2 + 2 + 2 + 2 + 2 + 2$.

(b) Prime factorization: $16 =$ _____ (How many 2s multiplied?)

3. [Square Numbers Verification] Which of the following numbers are square numbers? Circle them and write their twin factors:

6: Square? Yes / No

9: Square? Yes / No

18: Square? Yes / No

25: Square? Yes / No

36: Square? Yes / No

72: Square? Yes / No

Explain why 6 is NOT a square number:

Section 4: Distributive Property vs Pure Multiplication (7 pts each, 28 pts) **Score: _____**

4. [Distributive Law Diagnosis] In which of the following can you use the distributive property? State YES or NO and explain:

(a) $4 \times (2.5 + 7.5)$

Distributive Property applicable? _____

(b) $4 \times (2.5 \times 7.5)$

Distributive Property applicable? _____

Explanation:

5. [Two-Step Word Problem] A warehouse stores boxes in stacks. Each stack has 3^2 layers, and each layer contains 2^3 boxes.

(a) Calculate how many boxes are in one stack.

(b) If there are 5 identical stacks, find the total number of boxes.

(a) $3^2 =$ _____, $2^3 =$ _____.

Boxes in one stack = $3^2 \times 2^3 =$ _____.

(b) Total boxes in 5 stacks = _____.



[Foundational Clinic 2] Answer Key & Scoring Guide

Section 1: Expanded Form Table (24 pts)

- $2^4 = 2 \times 2 \times 2 \times 2 = 16$
- $3^3 = 3 \times 3 \times 3 = 27$
- $3^4 = 3 \times 3 \times 3 \times 3 = 81$ (NOT 12)
- $4^2 = 4 \times 4 = 16$
- $5^3 = 5 \times 5 \times 5 = 125$
- $10^3 = 10 \times 10 \times 10 = 1000$

Section 2: Exponents Product Rule (24 pts)

1. (a) $10^{4+5} = 10^9$ (NOT 10^{20}); (b) $2^{3+4} = 2^7 = 128$; (c) $3^{2+5} = 3^7$; (d) $10^0 = 1$; (e) $5^{1+2+3} = 5^6$; (f) $\frac{1}{10^3} = 0.001$.

Section 3: Factors, Multiples & Squares (24 pts)

2. (a) Addition; (b) $16 = 2 \times 2 \times 2 \times 2 = 2^4$ (four 2s multiplied).
3. Squares: 9 (3×3), 25 (5×5), 36 (6×6). Not squares: 6, 18, 72. 6 is not a square because its prime factors 2×3 are different.

Section 4: Distributive Property & Applications (28 pts)

4. (a) YES: $4 \times 2.5 + 4 \times 7.5 = 10 + 30 = 40$. (b) NO: $4 \times 2.5 \times 7.5 = 10 \times 7.5 = 75$.
5. (a) $3^2 = 9$, $2^3 = 8$. One stack = $9 \times 8 = 72$ boxes. (b) 5 stacks = $72 \times 5 = 360$ boxes.